NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

Department of Cyber Security

Mobile Forensics (MBF712S) Programme: 07BCCY Bachelor of Computer Science in Cyber Security

Lab 1

Facilitator: Ms. Hillary Kwala & Mr. Immanuel Nampala

Moderator: Professor Michael Olusanya

CLASS GROUP#: **DATE:** 2026 **MARK:** /30 .....%

GROUP MEMBERS

	Surname	**Initials**	**Student number**	**Signature**
1.				(Leader)
2.				
3.				

Objective(s):

The objective of this lab is to show evidence of student ability in mobile device evidence extraction and analysis. Through the use of various software and hardware architectures, recovery methods, techniques and tools.

Instructions:

1. This lab is based on "Practical Mobile: Forensically Investigate and Analyse iOS, Android and Windows 10 Devices Fourth Edition"
2. You should form your practical group of a minimum of 2 students and a maximum of 3 students **from the same mode of study.** **(NO INDIVIDUAL WORK WILL BE ACCEPTED!)** You are to notify the course facilitator about your groups.
3. Use the lab template provided for reporting. **Convert your report to pdf before uploading.** No email submissions will be accepted, pay attention to deadlines. E.g. (Lab1_Surname_FT-1)
4. If your report includes more than one file, zip before submission.

5. Your report should always include a section stipulating the role of each group member clearly without ambiguity.
6. Special tool: one iphone or ipad

Background:

libimobiledevice is a cross-platform software library that talks the protocols to interact with iOS devices. Unlike other projects, it does not depend on using any existing proprietary libraries and does not require jailbreaking. Access filesystem of a device, access documents of file sharing apps, retrieve information about a device and modify various settings, backup and restore the device in a native way compatible with iTunes. Manage app icons arrangement on the device, install, remove, list and basically manage apps. Activate a device using official servers, manage contacts, calendars, notes and bookmarks, retrieve and remove crashreports. Retrieve various diagnostics information, establish a debug connection for app debugging, mount filesystem images, forward device notifications, manage device provisioning. Take screenshots from the device screen (requires mounted developer image). Simulate changed geolocation of the device (requires mounted developer image).

Task: Identifying the correct hardware model

To obtain the iPhone model and its iOS version information on a Windows 10 workstation, follow these steps:

1. Download the latest binaries from the following link: https://github.com/iFred09/libimobiledevice-windows
2. Unzip the libimobiledevice-windows-master
3. Open command prompt (cmd) and change the directory to the one with binaries (use the **cd** command)
4. Connect the iPhone on your workstation using a USB/type-c cable (for the newer iOS versions, the password is required), and run the ideviceinfo command with the -s option, as shown in the following code: **ideviceinfo -s**
5. What is your final result?

   Show all important screenshots and details. Your screenshots should demonstrate that they are from your own workstation and not generic images copied from other sources.

   a) Why developing an understanding of the underlying components of a mobile device is important?
   b) Propose alternative method(s)/technique(s) for achieving similar results.

Cover Page

Mobile Forensics (MBF 712S) Lab Investigation	
Lab No:	*Lab 1*
Lead Investigators:	*Barnabas iihuhwa 223081701* *Titus Shaangelo 222084774*
Qualifications:	Bachelor of Computer Science in Cyber Security (Undergraduate)
Nature of Case: (Lab Objective)	The objective of this lab is to demonstrate the ability to identify and analyse a mobile device using forensic tools. The lab focuses on extracting device information such as the hardware model and iOS version without modifying the device data.
Due Date:	*02 March 2026*

Description of Evidence		**Vendor Name**	**Model /Serial No.**
Item #1	iPhone device	Apple Inc	iPhone 15 pro max
Item #2	USB / Type-C data cable	Apple	USB-C
Item #3	Windows 10 workstation	HP	HP 250 15.6-inch G9 notebook PC (7N0M6ES)
Item #4	libimobiledevice-windows binaries	libimobiledevice (Open Source)	Version [e.g., 1.3.0
Item #5	Command Prompt (CMD)	Microsoft	Windows 10 Build

1. Introduction

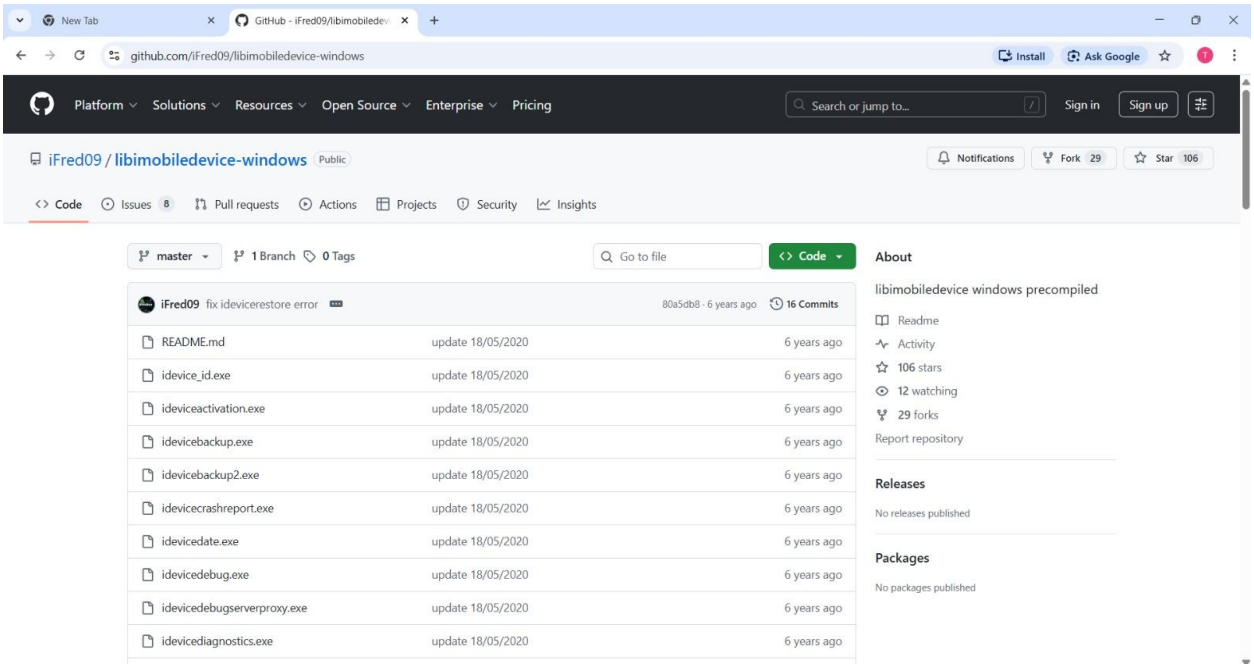
- Mobile device forensics focuses on extracting and analyzing digital evidence from smartphones in a way that preserves its integrity and admissibility. It is based on forensic principles such as maintaining a clear chain of custody, preventing data alteration, and using validated tools and commands to access device information. In this lab, the investigation relied on establishing a trusted connection between the computer and the mobile device, identifying the device, and retrieving system data using command-line forensic utilities. These processes demonstrate how investigators can safely obtain key information such as device identity, software version, and configuration details, which can later support an investigation or be used as evidence in a legal context.

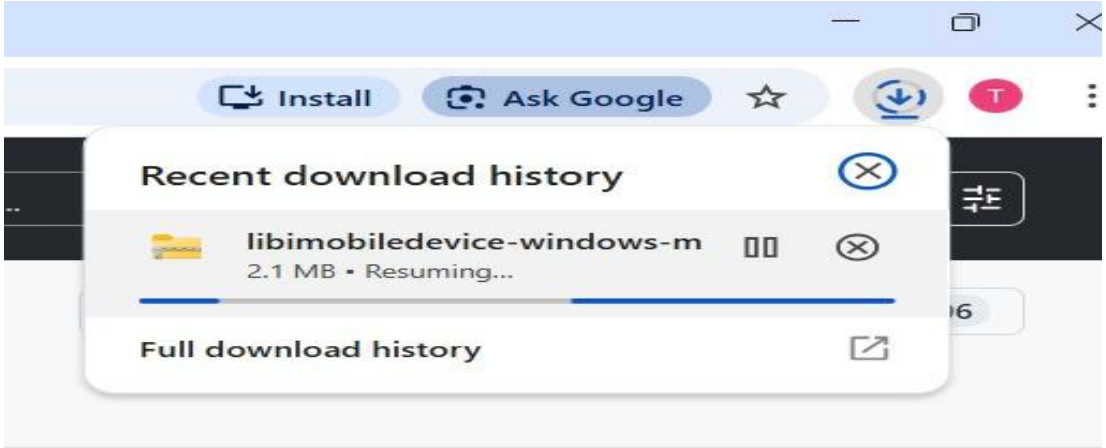
2. Lab Procedures

Tool Acquisition and Preparation

- The libimobiledevice-windows binaries were downloaded from the official repository and extracted to a local directory on a Windows 10 workstation to prepare for device communication.

Screenshot 1: Downloaded and extracted libimobiledevice-windows directory on the workstation.

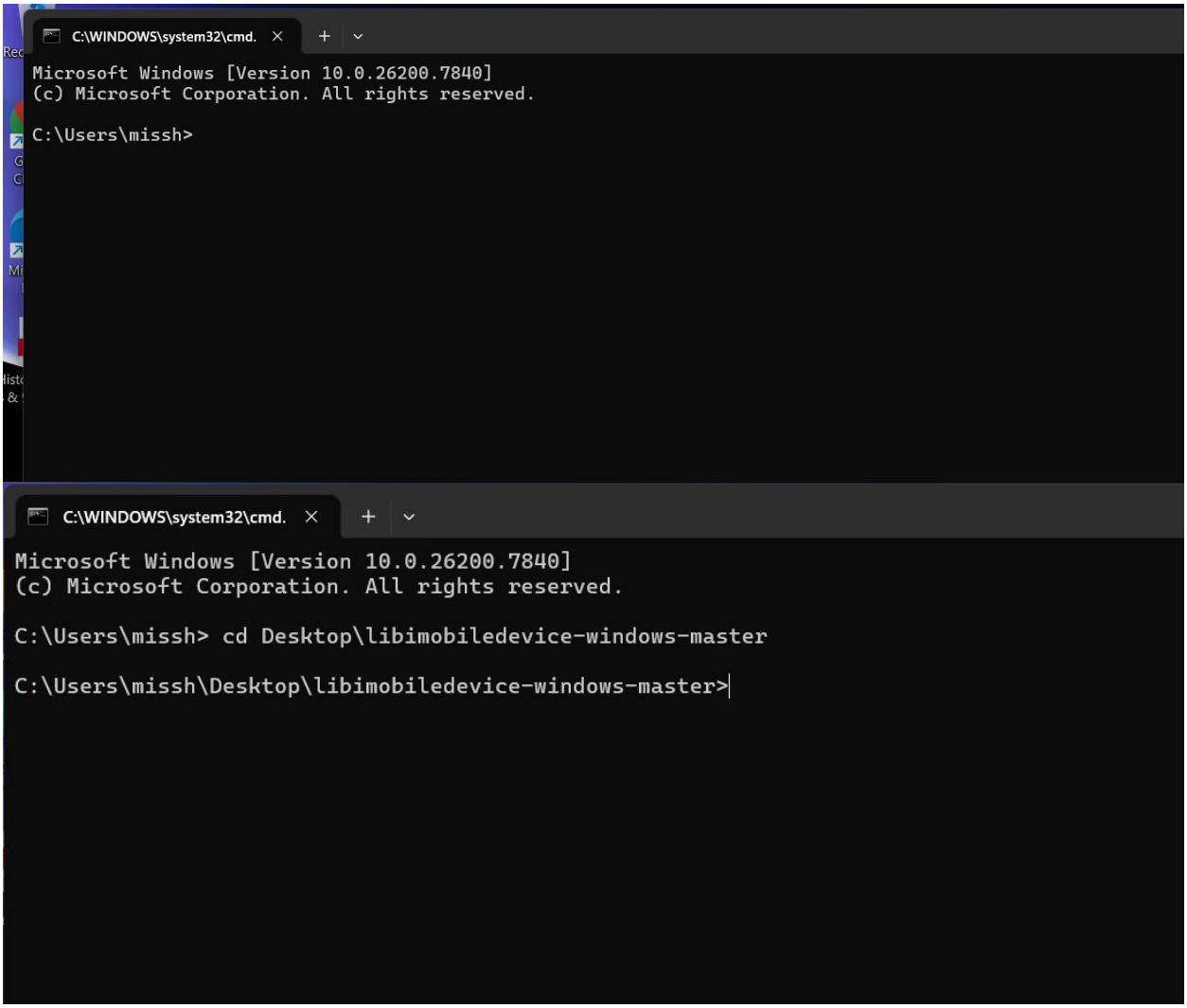




{{ıt

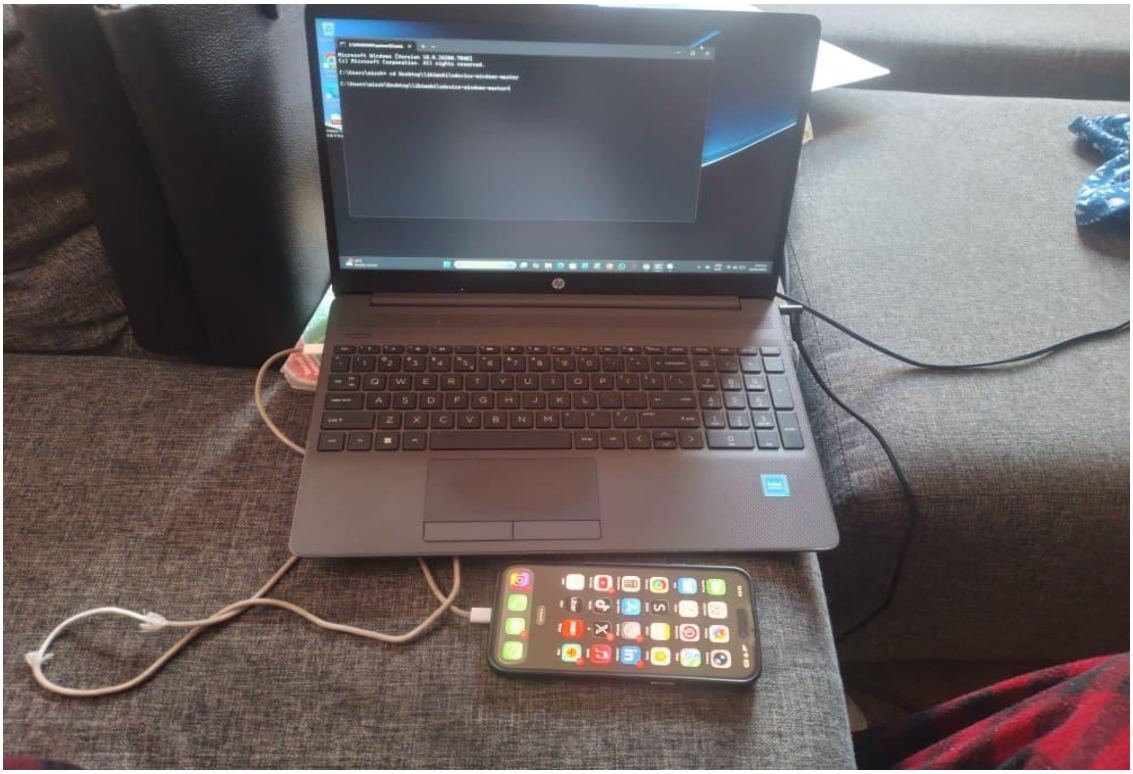
- **Command Line Environment Setup**
  Command Prompt (CMD) was launched, and the working directory was changed to the folder containing the extracted libimobiledevice binaries using the `cd` command.

Screenshot 2: Command Prompt showing directory navigation to the libimobiledevice binaries



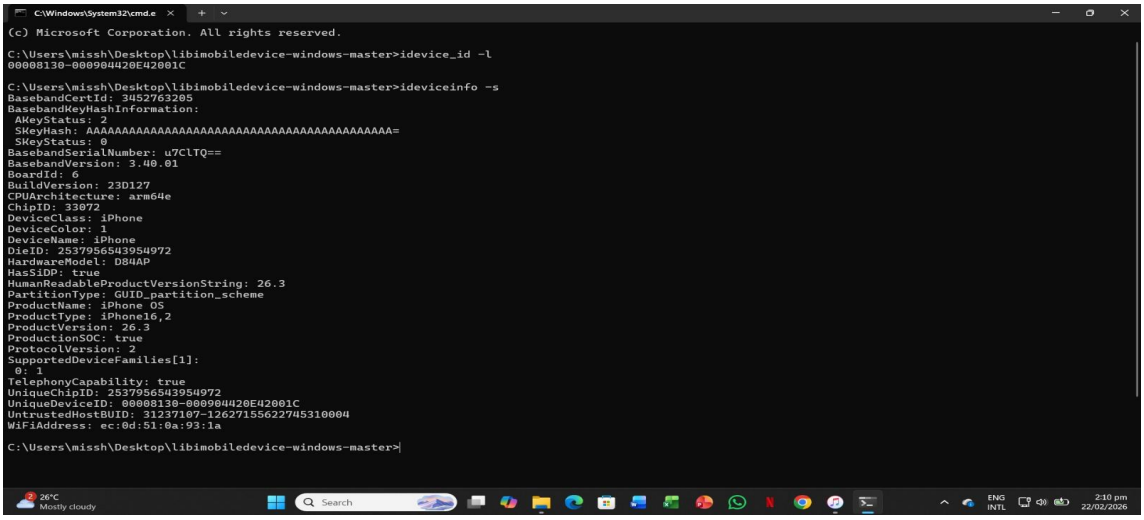
- **Device Connection and Authorization**
  The iPhone was connected to the workstation via USB. The device was unlocked, and the “Trust This Computer” prompt was accepted to enable authorised communication between the device and workstation.

Screenshot 3: iPhone device connected and successfully recognized by the workstation.



- **Device Information Extraction**
  The command ideviceinfo -s was executed in Command Prompt to retrieve system-level information from the connected iPhone.

Screenshot 4: Execution of the ideviceinfo -s *command in Command Prompt.*

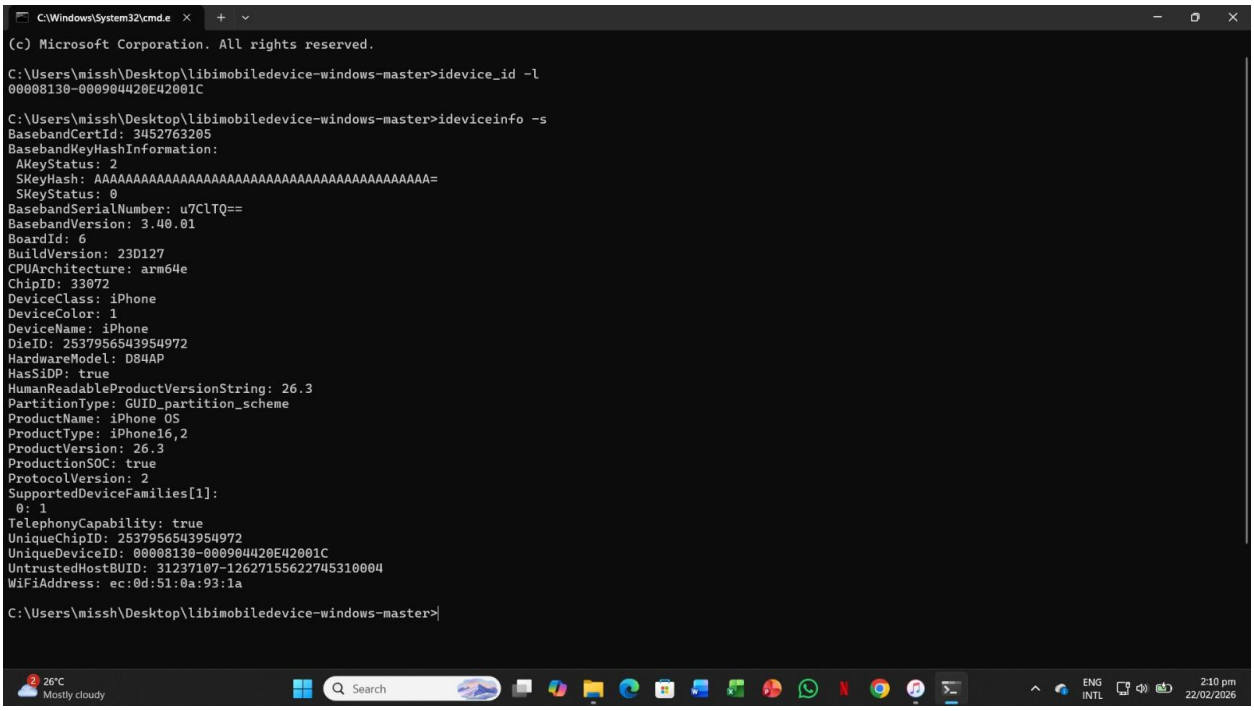


```
C:\Windows\System32\cmd.e
(c) Microsoft Corporation. All rights reserved.

C:\Users\missh\Desktop\libimobiledevice-windows-master>idevice_id -l
00008130-000904420E42001C

C:\Users\missh\Desktop\libimobiledevice-windows-master>ideviceinfo -s
BasebandCertId: 3452763205
BasebandKeyHashInformation:
 AKeyStatus: 2
 SKeyHash: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA=
 SKeyStatus: 0
BasebandSerialNumber: u7ClTQ==
BasebandVersion: 3.40.01
BoardId: 6
BuildVersion: 23D127
CPUArchitecture: arm64e
ChipID: 33072
DeviceClass: iPhone
DeviceColor: 1
DeviceName: iPhone
DieID: 25379565439549 72
HardwareModel: D84AP
HasSiDP: true
HumanReadableProductVersionString: 26.3
PartitionType: GUID_partition_scheme
ProductName: iPhone OS
ProductType: iPhone16,2
ProductVersion: 26.3
ProductionSOC: true
ProtocolVersion: 2
SupportedDeviceFamilies[1]:
 0: 1
TelephonyCapability: true
UniqueChipID: 25379565439549 72
UniqueDeviceID: 00008130-000904420E42001C
UntrustedHostBUID: 31237107-1262715562274531 0004
WiFiAddress: ec:0d:51:0a:93:1a

C:\Users\missh\Desktop\libimobiledevice-windows-master>
```

Results Obtained

The command returned detailed system information, including the iPhone hardware model and installed iOS version. This confirmed successful communication between the forensic tool and the mobile device.

Screenshot 5: Output of the ideviceinfo -s *command displaying the iPhone model and iOS version.*



```
C:\Windows\System32\cmd.e

(c) Microsoft Corporation. All rights reserved.

C:\Users\missh\Desktop\libimobiledevice-windows-master>idevice_id -l
00008130-000904420E42001C

C:\Users\missh\Desktop\libimobiledevice-windows-master>ideviceinfo -s
BasebandCertId: 3452763205
BasebandKeyHashInformation:
 AKeyStatus: 2
 SKeyHash: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA=
 SKeyStatus: 0
BasebandSerialNumber: u7ClTQ==
BasebandVersion: 3.40.01
BoardId: 6
BuildVersion: 23D127
CPUArchitecture: arm64e
ChipID: 33072
DeviceClass: iPhone
DeviceColor: 1
DeviceName: iPhone
DieID: 2537956543954972
HardwareModel: D84AP
HasSiDP: true
HumanReadableProductVersionString: 26.3
PartitionType: GUID_partition_scheme
ProductName: iPhone OS
ProductType: iPhone16,2
ProductVersion: 26.3
ProductionSOC: true
ProtocolVersion: 2
SupportedDeviceFamilies[1]:
 0: 1
TelephonyCapability: true
UniqueChipID: 2537956543954972
UniqueDeviceID: 00008130-000904420E42001C
UntrustedHostBUID: 31237107-1262715562274531000 4
WiFiAddress: ec:0d:51:0a:93:1a

C:\Users\missh\Desktop\libimobiledevice-windows-master>
```

3. Lab Challenges/Problems Encountered

No major challenges were encountered during the laboratory exercise. Initial communication required the iPhone device to be unlocked and trust permission to be granted before data extraction could proceed.

4. Conclusion

The lab successfully demonstrated the practical process of mobile device evidence identification and extraction using forensic tools on an iPhone. Through the use of command-line utilities, it was possible to establish a connection with the device and retrieve important system information such as the device name, model, and iOS version. The activity highlighted the importance of correct procedures, proper driver installation, and establishing trust between the device and workstation in order to access forensic data. It also showed that challenges such as device

detection errors can occur, but can be resolved through proper troubleshooting steps. Overall, the lab provided a clear understanding of how mobile forensic investigations are conducted and the importance of following sound forensic practices to preserve and analyze digital evidence.

5. The role played by each group member:

Group Member 1 (Leader): Barnabas iihuhwa
Connected the iPhone device, executed the ideviceinfo command, and captured all required screenshots.
Group Member 2: Titus shaangelo
Compiled the laboratory report, documented procedures, and reviewed findings for accuracy and completeness.

References

1. libimobiledevice. (2024). *libimobiledevice: Cross-platform software protocol library to communicate with iOS devices*. GitHub. https://github.com/libimobiledevice/libimobiledevice
2. Microsoft Corporation. (2023). *Windows command-line reference*. https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/windows-commands
3. Apple Inc. (2023). *iTunes User Guide*. https://support.apple.com/guide/itunes/welcome/mac